Converting Markdown to Other Formats with knitr::pandoc()

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A bit *introduction* here.

1 Start with a cool section

You can use traditional **Markdown** syntax, such as links and `code`.

2 Followed by another section

Of course you can write lists:

- apple
- pear
- banana

Or ordered lists:

1. items
2. will
3. be
4. ordered
    - nested
    - items

3 More sections

3.1 Hi

hi hi

3.2 Hello

hello hello

3.3 Howdy

howdy howdy

4 Okay, some R code

```
fit = lm(dist ~ speed, data = cars)
b = coef(fit)  # coefficients
summary(fit)
```

```
##
## Call:
## lm(formula = dist ~ speed, data = cars)
##
## Residuals:
##    Min     1Q Median     3Q    Max
```

```
## -29.07  -9.53  -2.27   9.21  43.20
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -17.579      6.758   -2.60    0.012 *
## speed          3.932      0.416    9.46  1.5e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 15.4 on 48 degrees of freedom
## Multiple R-squared:  0.651,  Adjusted R-squared:  0.644
## F-statistic: 89.6 on 1 and 48 DF,  p-value: 1.49e-12
```

The code will be highlighted in all output formats.

5 And some pictures

```
par(mfrow = c(2, 2), pch = 20, mar = c(4, 4, 2, .1), bg = 'white')
plot(fit)
```

6 A little bit math

Our regression equation is $Y = -17.5791 + 3.9324x$, and the model is:

$$Y = \beta_0 + \beta_1 x + \epsilon$$

7 Pandoc extension: definition lists

Programmer A programmer is the one who turns coffee into code. LaTeX
: A simple tool which is nothing but a couple of backslashes.

8 Pandoc extension: examples

We have some examples.

(1) Think what is `0.3 + 0.4 - 0.7`. Zero. Easy.
(2) Now think what is `0.3 - 0.7 + 0.4`. Still zero?

People are often surprised by (2).

9 Pandoc extension: tables

A table here.

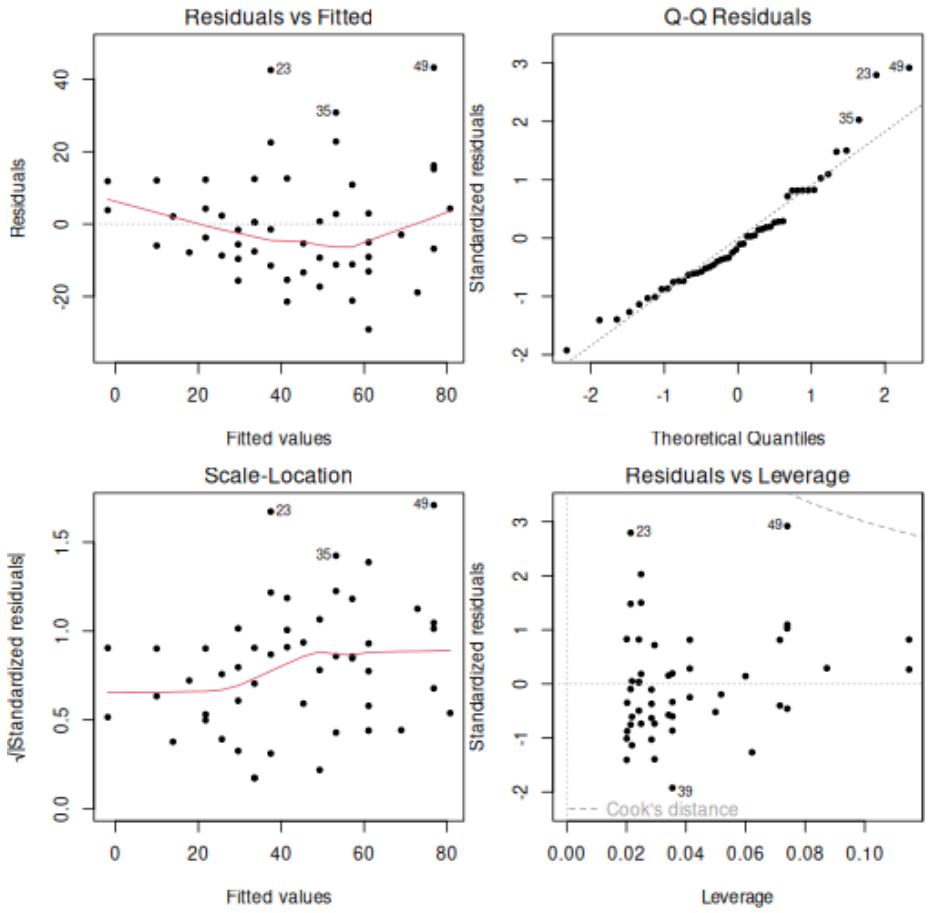


Figure 1: Regression diagnostics

Table 1: Demonstration of simple table syntax.

Right	Left	Center	Default
12	12	12	12
123	123	123	123
1	1	1	1

10 Pandoc extension: footnotes

We can also write footnotes[1].

Or write some inline footnotes[2].

11 Pandoc extension: citations

We compile the R Markdown file to Markdown through **knitr** [@xie2013] in R [@R-base]. For more about @xie2013, see http://yihui.org/knitr.

12 References

[1] hi, I'm a footnote

[2] as you can see here